

Learning with AI

Web Dashboard Development

Project: Mold Growth Risk Predictor

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What I Learned

- How to create a web server using ESP32
- How to display real-time data in a browser
- How to use HTML, CSS, and JavaScript together
- How to visualize data with a graph

Why This is Important

- Data alone is not useful without interpretation
- A dashboard helps users understand conditions
- Supports identifying mold risk over time

What I Built

- Web dashboard accessible via ESP32 IP
- Displays:
 - Temperature
 - Humidity
 - Mold risk (Low, Moderate, High)
 - Trend (Rising, Falling, Stable)

Data Visualization

- Stored recent readings
- Built a humidity graph using HTML canvas
- Shows how conditions change over time

UI Improvements

- Designed a clean layout
- Added color-coded risk levels
- Used consistent color palette
- Organized dashboard into sections

How AI Helped Me

- Generated and explained HTML/CSS/JS code
- Helped debug styling issues
- Guided creation of graph visualization
- Suggested UI improvements

Challenges & Solutions

- CSS not applying → fixed HTML structure
- Page not updating → refreshed and re-uploaded
- Graph implementation → used canvas with AI guidance

Reflection

- Learned how embedded systems connect to web interfaces
- Built a complete pipeline: sensor → data → dashboard
- Improved both technical and design skills